Ex5b: Information Extraction using spaCy

This notebook demonstrates Information Extraction using spaCy on the Amazon Fine Food Reviews Dataset.

```
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!pip install -q kaggle spacy nltk pandas
!python -m spacy download en_core_web_sm

import os
import re
import pandas as pd
import nltk
import spacy


nltk.download('stopwords')
nltk.download('punkt')
from nltk.corpus import stopwords
```

```
Collecting en-core-web-sm==3.8.0
  Downloading en_core_web_sm-3.8.0-py3-none-any.whl (12.8 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 41.6 MB/s  0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
```

To download the dataset from Kaggle, make sure you have uploaded your `kaggle.json` API token to the environment. If you are running this in Colab, you can upload it using the file explorer on the left or set your environment variables.

```
# 3. Download and Load the Dataset
# Setting up Kaggle API credentials (make sure to upload kaggle.json to ~/.kaggle/ or /content/)
if os.path.exists('kaggle.json'):
    !mkdir -p ~/.kaggle
    !cp kaggle.json ~/.kaggle/
    !chmod 600 ~/.kaggle/kaggle.json

# Download dataset
!kaggle datasets download -d snap/amazon-fine-food-reviews -f Reviews.csv --unzip

# Load the Dataset into DataFrame
df = pd.read_csv('Reviews.csv')

# a. Select the column containing review text
# b. Remove any missing or null entries
df_clean = df[['Text']].dropna()
```

```
# c. Limit the dataset to 10,000 reviews for faster processing
df_subset = df_clean.head(10000).copy()
display(df_subset.head())
```

```
Warning: Looks like you're using an outdated `kaggle` version (installed: 2.0.2), please consider
Dataset URL: https://www.kaggle.com/datasets/snap/amazon-fine-food-reviews
License(s): CC0-1.0
Downloading Reviews.csv.zip to /content
100% 115M/115M [00:03<00:00, 33.8MB/s]

---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call last)
/tmp/ipykernel_10826/1769643120.py in <cell line: 0>()
     10
     11 # Load the Dataset into DataFrame
---> 12 df = pd.read_csv('Reviews.csv')
     13
     14 # a. Select the column containing review text

 4 frames
/usr/local/lib/python3.13/dist-packages/pandas/io/common.py in get_handle(path_or_buf, mode,
encoding, compression, memory_map, is_text, errors, storage_options)
    871         if ioargs.encoding and "b" not in ioargs.mode:
    872             # Encoding
--> 873             handle = open(
    874                 handle,
    875                 ioargs.mode,

FileNotFoundError: [Errno 2] No such file or directory: 'Reviews.csv'
```

Next steps: Explain error

```
# 4. Preprocessing step
# a. Lowercase
# b. Remove punctuation

def preprocess_text(text):
    text = text.lower()
    text = re.sub(r'[^\w\s]', '', text) # Remove punctuation
    return text

df_subset['Cleaned_Text'] = df_subset['Text'].apply(preprocess_text)
display(df_subset[['Text', 'Cleaned_Text']].head())
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_10826/1536096591.py in <cell line: 0>()
      8     return text
      9
---> 10 df_subset['Cleaned_Text'] = df_subset['Text'].apply(preprocess_text)
     11 display(df_subset[['Text', 'Cleaned_Text']].head())

NameError: name 'df_subset' is not defined
```

Next steps: Explain error

Gemini

```
# Load spacy model
nlp = spacy.load('en_core_web_sm')
stop_words = set(stopwords.words('english'))

# 5. Tokenization, Stopwords removal, POS Tagging, and NER extraction
def extract_information(text):
    doc = nlp(text)

    # Tokenization, stopword removal, and non-alphabetic filtering
    tokens = [token.text for token in doc if token.text not in stop_words and token.is_alpha]

    # Re-process filtered tokens with spaCy to get POS tags
    filtered_doc = nlp(" ".join(tokens))

    # POS Tagging
    pos_tags = [(token.text, token.pos_) for token in filtered_doc]

    # Named Entity Recognition (NER)
    entities = [(ent.text, ent.label_) for ent in filtered_doc.ents]

    return tokens, pos_tags, entities

# Apply to a sample of 10 reviews first to demonstrate functionality
sample_df = df_subset.head(10).copy()
sample_results = sample_df['Cleaned_Text'].apply(extract_information)

sample_df['Tokens'] = [res[0] for res in sample_results]
sample_df['POS_Tags'] = [res[1] for res in sample_results]
sample_df['Entities'] = [res[2] for res in sample_results]

display(sample_df[['Cleaned_Text', 'Tokens', 'POS_Tags', 'Entities']].head(5))
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_10826/435661460.py in <cell line: 0>()
     22
     23 # Apply to a sample of 10 reviews first to demonstrate functionality
---> 24 sample_df = df_subset.head(10).copy()
     25 sample_results = sample_df['Cleaned_Text'].apply(extract_information)
     26

NameError: name 'df_subset' is not defined
```

Next steps: Explain error